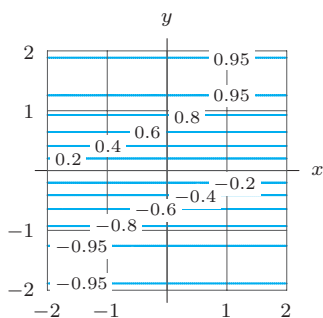
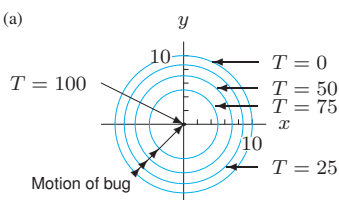




- 13 $2x^2 + y^2 = k$
 15 $(x-1)^2 + (y-2)^2 + (z-3)^2 = 25$
 17 $(-2, 3, -6); 7$
 19 Linear
 21 $f(x, y, z) = x^2 + y^2$
 23 $z = f(x, y) = 4 - 2x - 4y/3$
 $g(x, y, z) = (x/2) + (y/3) + (z/4) = 1$
 25 $z = f(x, y) = -\sqrt{4 - (x-3)^2 - y^2}$
 $g(x, y, z) = (x-3)^2 + y^2 + z^2 = 4$
 27 Cylinder
 29 $g(x, y) = 200x + 100y$
 31 Doubles production
 37 (a) No
 (b) No
 (c) No
 (e) 0.5 of GPA
 39 (a)



- (b) Toward origin
 41 (a)
 (b)
 (c)
 47 (a) $9 + 3x + 4y, 21 + 7x + 8y$
 (b) $189 + 63x + 64y$

Section 13.1

- 1 $\vec{a} = \vec{i} + 3\vec{j}$
 $\vec{b} = 3\vec{i} + 2\vec{j}$
 $\vec{v} = -2\vec{i} - 2\vec{j}$
 $\vec{w} = -\vec{i} + 2\vec{j}$
 3 $-3\vec{i} - 4\vec{j}$
 5 $\vec{a} = \vec{b} = \vec{c} = 3\vec{k}$
 $\vec{d} = 2\vec{i} + 3\vec{k}$
 $\vec{e} = \vec{j}$
 $\vec{f} = -2\vec{i}$
 7 $\vec{i} + 3\vec{j}$
 9 $-4.5\vec{i} + 8\vec{j} + 0.5\vec{k}$
 11 $-3\vec{i} - 12\vec{j} + 3\vec{k}$
 13 $0.9\vec{i} + 0.2\vec{j} - 0.7\vec{k}$
 15 $\sqrt{6}$
 17 $\sqrt{11}$
 19 5.6
 21 $-6\vec{i} + 20\vec{j} + 13\vec{k}$
 23 $21\vec{j}$
 25 $2\sqrt{73}$
 27 $0.6\vec{i} - 0.8\vec{k}$
 29 $-\vec{i}/2 + \vec{j}/4 + \sqrt{11}\vec{k}/4$
 31 0, -10
 33 $-\frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j}$
 35 (a) $\vec{u} = 8.2\vec{i} + 1.8\vec{j}$,
 $\vec{v} = -8.2\vec{i} - 1.8\vec{j}$
 (b) $\vec{v} = -\vec{u}$
 37 (a) True
 (b) Does not make sense
 (c) True
 (d) False
 39 (a) $t = 1$
 (b) No t values
 (c) Any t values
 41 $-\vec{i}/2 + \sqrt{3}\vec{j}/2$
 $-\vec{i}/2 - \sqrt{3}\vec{j}/2$
 45 Opposite direction if $c < 0$, no direction if $c = 0$
 47 $\vec{w} = \vec{0}$ so $\vec{v} + \vec{w} = \vec{v}$
 49 $\vec{u} = \vec{i}, \vec{v} = (1/2)\vec{i} + (\sqrt{3}/2)\vec{j}$
 51 False
 53 True
 55 False
 57 False
 59 False

Section 13.2

- 1 Scalar
 3 Scalar
 5 Vector
 7 $-37.59\vec{i}, -13.68\vec{j}$
 9 $21\vec{i} + 35\vec{j}$
 11 (a) 8.64 km/hr
 (b) 0.093 radian or about 5° off course
 13 (a) $17.93\vec{i} - 7.07\vec{j}$
 (b) 19.27 km/hr
 (c) 21.52° south of east
 15 48.3° east of north
 744 km/hr
 17 4.87° north of east
 540.63 km/hr

- 19 $\vec{E} = 27.908\vec{i} + 1.823\vec{j}$
 Speed 27.97 km/hr in direction 3.74° north of east
 21 38.7° south of east
 23 $-98.76\vec{i} + 18.94\vec{j} + 2998.31\vec{k}$
 2998.31 newtons directly up
 25 $0.4v\vec{i} + 0.7v\vec{j}$
 27 $0.01875\vec{i} + .0125\vec{j} + .03125\vec{k}$
 39 $\vec{i}$ or $\vec{j}$ components could be different
 41 $\vec{F} = \vec{i} - 2\vec{j}$
 43 Yes
 45 Yes
 47 Yes

Section 13.3

- 1 -14
 3 14
 5 -2
 7 $28\vec{j} + 14\vec{k}$
 9 185
 11 $\vec{i} + 3\vec{j} + 2\vec{k}$ (multiples of)
 13 $3\vec{i} + 4\vec{j} - \vec{k}$ (multiples of)
 15 $3x - y + 4z = 6$
 17 $x - y + z = 3$
 19 $2x + 4y - 3z = 5$
 21 $2x - 3y + 5z = -17$
 23 $2\pi/3$ radians (120°)
 25 $\pi/6$ radians (30°)
 27 (a) $(2/\sqrt{13})\vec{i} + (3/\sqrt{13})\vec{j}$
 (b) Multiples of $3\vec{i} - 2\vec{j}$
 29 (a) $(21/5, 0, 0)$
 (b) $(0, -21, 0); (0, 0, 3)$ (for example)
 (c) $\vec{n} = 5\vec{i} - \vec{j} + 7\vec{k}$ (for example)
 (d) $21\vec{j} + 3\vec{k}$ (for example)
 31 Possible answers:
 (a) $2\vec{i} + 3\vec{j} - \vec{k}$
 (b) $3\vec{i} - 2\vec{j}$
 33 (a) $(\vec{a}, \vec{c})$ and $(\vec{b}, \vec{d})$
 (b) None
 (c) $(\vec{a}, \vec{d})$ and $(\vec{c}, \vec{d})$
 (d) $(\vec{a}, \vec{b})$ and $(\vec{b}, \vec{c})$
 35 Many answers possible
 (a) $8\vec{i} + 6\vec{j}$
 (b) $-3\vec{i} + 4\vec{j}$
 37 (a) Increases
 (b) Decreases
 39 71.57°
 41 (a) $\sqrt{2}$
 (b) 0.32
 43 (a) $\vec{F}_{\text{parallel}} = -0.168\vec{i} - 0.224\vec{j}$
 (b) $\vec{F}_{\text{perp}} = 0.368\vec{i} - 0.276\vec{j}$
 (c) $W = -1.4$
 45 (a) $\vec{F}_{\text{parallel}} = \vec{0}$
 (b) $\vec{F}_{\text{perp}} = \vec{F}$
 (c) $W = 0$
 47 (a) $\vec{F}_{\text{parallel}} = \vec{F}$